CE8014 - Agentic AI: Foundations and Development - Assignment 02

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1 Implementation Logic

1.1 System Prompt Strategy

The system prompt instructs the model on the exact output format expected at every step, enforces the ReAct loop, and provides a one-shot in-context learning example. The full prompt is described below.

```
You are a ReAct agent that answers questions by alternating between
Thinking, Acting, and Observing.

Always follow this EXACT format:

Thought: <your reasoning about what to do next>
Action: <tool name - must be one of: Search>
Action Input: <query or input for the tool>
Observation: <result returned by the tool - provided by the system,
              never write this yourself>
... (repeat Thought/Action/Action Input/Observation as needed)
Thought: <your final reasoning once you have enough information>
Final Answer: <your complete, direct answer to the original question>

Rules:
- You MUST begin every response with "Thought:".
- NEVER write an "Observation:" line yourself - the system injects it
  after each Action.
- If a search returns empty or irrelevant results, reflect on why and
  try a differently phrased query.
- For multi-part questions, decompose into sequential searches before
  computing the answer.
- You have at most 5 steps total. Be concise and efficient.

=== EXAMPLE ===
Question: What is the population of Australia?

Thought: The user wants Australia's current population. I will search
         for the latest figure.
Action: Search
Action Input: Australia population 2025
Observation: Australia's population is approximately 26.8 million as of
             2025, according to the Australian Bureau of Statistics.

Thought: I have the answer I need.
Final Answer: Australia's population is approximately 26.8 million
              as of 2025.
=== END EXAMPLE ===

Now answer the following question.
```

Listing 1: Full System Prompt (agent.py)

Highlighted: The Few-Shot (One-Shot) Example

The block between `=== EXAMPLE ===` and `=== END EXAMPLE ===` is the one-shot demonstration. It shows the model:

- That every response must open with `Thought:`.

- The exact syntax of `Action:` and `Action Input:`.
- That `Observation:` is injected externally and must never be self-written.
- How to transition from a satisfied observation to a `Final Answer:`.

Why this matters (In-Context Learning): Without a concrete example, LLMs frequently deviate from the required format. Specifically, they may skip the `Thought:` prefix or combine `Action Input` and a full sentence. Furthermore, they can even hallucinate an `Observation:` and result before the system can inject the real one. A complete example can "teach" the model how to reason in a proper and reliable way, resulting in the stability of the model's behavior.

1.2 The Loop Mechanism

The ReAct loop is implemented in `ReActAgent.execute()` in `agent.py`. Specifically, for each iteration, **conversation history**, which is maintained as a flat list of strings, is concatenated and passed back to the LLM. This allows the model to read full prior thoughts, actions, and observations within one context window.

As illustrated in Figure 1, the loop proceeds as follows:

1. **Initialise history** with the raw question: `self.messages = [f"Question: {query}\n\n"]`.
2. **Call LLM** via `_call_llm()`. The method sends the accumulated `"".join(self.messages)` as the `user` turn, with the system prompt in the `system` role, and sets `stop=["\nObservation:"]`. The stop sequence prevents the model from generating its own observation.
3. **Append LLM output** to `self.messages`.
4. **Check for `Final Answer`**. If the regex `r"^Final Answer:\s*(.+)"` matches, return immediately.
5. **Parse Action and dispatch tool**. `_parse_action()` extracts the tool name and input via regex. The tool is looked up in the `TOOLS` dict and called. Unknown tools or missing actions produce a descriptive error string rather than crashing.
6. **Inject observation** back into history: `self.messages.append(f"\nObservation: {observation}\n")`. Because the entire `self.messages` list is concatenated on the next `_call_llm()` call, the model sees the observation as if it were part of a continuing monologue—which is exactly the "scratchpad" mechanism described in the ReAct paper [1].
7. **Hard iteration limit**: the loop runs at most `MAX_ITERATIONS = 5` times before returning a timeout message.

2 Benchmark Traces

2.1 Task 1 — Planning & Quantitative Reasoning

Question: *"What fraction of Japan's population is Taiwan's population as of 2025?"*

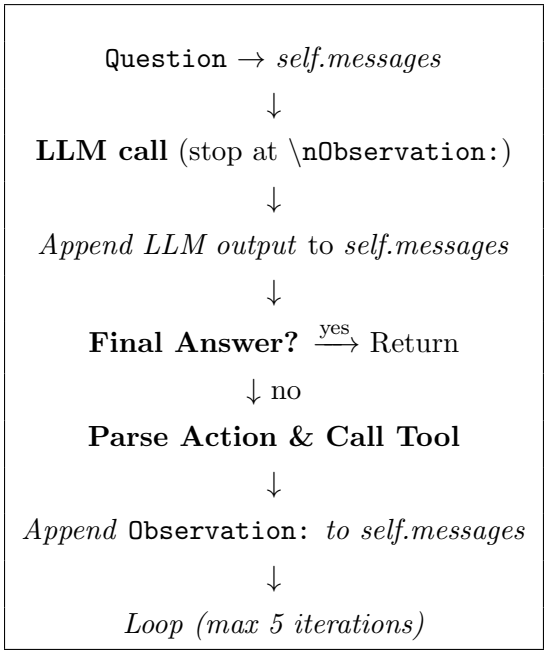

Figure 1: ReAct loop data-flow inside `ReActAgent.execute()`

Task Decomposition Analysis. As shown in Figure 2, the agent correctly **decomposed** the problem into two distinct search steps initially. In Step 1, instead of attempting to guess the ratio directly from training knowledge, it explicitly articulated the plan: *"I need to find both values. I'll start with Japan's 2025 population."*. Then, it took the action that called the tool to execute the plan and waited to receive the results exactly. After obtaining results, specifically population figures, it performed the division in Step 3 and emitted a `Final Answer`. This demonstrates the *planning* behaviour required by the rubric.

```
(agentic) mtoan65@mtoan65-Katana-15-B13VGK:~/Documents/NCU/ReAct_Agent$ python main.py

============================================================
Task 1: What fraction of Japan's population is Taiwan's population as of 2025?
============================================================

[Step 1]
Thought: To find the fraction of Japan's population that is Taiwan's population, I first need to find Japan's population as of 2025.
Action: Search
Action Input: Japan population 2025
Observation: Japan Population reached 123.2 million people in Dec 2025, compared with the previously reported figure of 123.7 million people in Dec 2024.
---
Population pyramid of Japan in 2025. Total population: 123103479. Interactive age and sex distribution visualization from 1950 to 2100.
---
## Japan Population (1950-2025). Population Economy Trade Health Education Development Labor Force Environment Crime Immigration Other. Population Growth Rate Density Urban Rural
Life Expectancy Birth Rate Death Rate Infant Mortality Rate Fertility Rate. ##### Total current population for Japan in 2025 was **121,960,408**, a **0.55% decline** from 2024. *
Total population for Japan in 2024 was **122,631,432**, a **1.51% decline** from 2023. * Total population for Japan in 2023 was **124,516,650**, a **0.49% decline** from 2022. * T
otal population for Japan in 2022 was **125,124,989**, a **0.44% decline** from 2021. Total population is based on the de facto definition of population, which counts all resident
s regardless of legal status or citizenship. The values shown are midyear estimates. The full historical dataset is available for download here: Population of Japan | Historical D
ata | 1950 - 2025.
```

(a) Step 1 — Agent searches for Japan's population (2025)

```
(agentic) mtoan65@mtoan65-Katana-15-B13VGK:~/Documents/NCU/ReAct_Agent$ python main.py

[Step 2]
Thought: I have found Japan's population as of 2025, which is approximately 121,960,408. Now, I need to find Taiwan's population as of 2025 to calculate the fraction.
Action: Search
Action Input: Taiwan population 2025
Observation: ## Taiwan Population (1950-2025). Population Economy Trade Health Education Development Labor Force Environment Crime Immigration Other. Population Growth Rate Densit
y Urban Rural Life Expectancy Birth Rate Death Rate Infant Mortality Rate Fertility Rate. ##### Total current population for Taiwan in 2025 was **23,974,031**, a **0.1% increase**
 from 2024. * Total population for Taiwan in 2024 was **23,950,214**, a **0.11% increase** from 2023. * Total population for Taiwan in 2023 was **23,923,276**, a **0.13% increase*
* from 2022. * Total population for Taiwan in 2022 was **23,893,394**, a **0.14% increase** from 2021. Total population is based on the de facto definition of population, which co
unts all residents regardless of legal status or citizenship. The values shown are midyear estimates. The full historical dataset is available for download here: Population of Tai
wan | Historical Data | 1950 - 2025.
---
## The total population in Taiwan was estimated at 23.3 million people in 2025, according to the latest census figures and projections from Trading Economics. Looking back, in the
 year of 1974, Taiwan had a population of 15.9 million people. ### Population in Taiwan is expected to reach 23.40 Million by the end of 2026, according to Trading Economics globa
l macro models and analysts expectations. In the long-term, the Taiwan Population is projected to trend around 23.50 Million in 2027, according to our econometric models. | Unempl
oyment Rate | 3.36 | 3.35 | percent | Jan 2026 |. | Wages | 76495.00 | 62459.00 | TWD/Month | Jan 2026 |. | Wages in Manufacturing | 81740.00 | 66575.00 | TWD/Month | Jan 2026 |.
Taiwan Export Orders Hit 6-Month Low. Taiwan Holds Rates as Expected. Taiwan Imports Growth at Over 1-Year Low. Taiwan Exports Growth Ease to 11-Month Low. Taiwan Trade Surplus at
 5-Month Low. Taiwan Inflation Rate Hits 10-Month High.
---
+ Senior Taiwan official attends MQ-9B delivery ceremony in U.S. + Senior Taiwan official attends MQ-9B delivery ceremony in U.S. + Kaohsiung mayor visits U.S. tech giants for AI,
 semiconductor cooperation. # Taiwan's population falls for 25th consecutive month. Taipei, Feb. 10 (CNA) Taiwan's population fell for a 25th consecutive month in January, with th
e number of live births falling by 304 from December 2025, according to data released Tuesday by the Ministry of the Interior (MOI). As of the end of January, Taiwan's population
totaled 23,289,045, down 10,087 from December 2025, the data showed. The MOI data also showed that the population aged 65 and older totaled 4,692,484, or 20.15 percent of the popu
lation, meeting the United Nations' definition of a "super-aged society," in which people 65 and older account for more than 20 percent of the population. ## Senior Taiwan officia
l attends MQ-9B delivery ceremony in U.S. 03/22/2026 02:04 PM. Click here to find out more about Focus Taiwan's privacy policy.

[Step 3]
Thought: I have found Taiwan's population as of 2025, which is approximately 23,974,031, but there's also a mention of 23.3 million and 23,289,045. For calculation purposes, I wil
l use the first figure as it seems to be a midyear estimate. Now, I can calculate the fraction of Japan's population that is Taiwan's population.
Action: None, I have all the necessary information.
Thought: To find the fraction, I will divide Taiwan's population by Japan's population. fraction = Taiwan's population / Japan's population = 23,974,031 / 121,960,408.
Final Answer: The fraction of Japan's population that is Taiwan's population as of 2025 is approximately 0.1965 or 19.65%.

>>> Final Answer: The fraction of Japan's population that is Taiwan's population as of 2025 is approximately 0.1965 or 19.65%.
```

(b) Steps 2 & 3 — Agent searches for Taiwan's population, then computes the fraction

Figure 2: Task 1 full console trace: Planning & Quantitative Reasoning

2.2 Task 2 — Technical Specificity

Question: *"Compare the main display specs of iPhone 15 and Samsung S24."*

Data Retrieval and Technical Specificity Analysis. The Figure 3 demonstrated that the agent found an answer about the main difference between the two phones. However, due to the difference in the retrieved data, as shown in the trace log, the agent did not point out the required difference. Although there are some differences between the required and actual answers, the actual answer still meets the requirements in another way. This proved that the agent worked correctly despite some changes in the fetched data.

```
(agentic) mtoan65@mtoan65-Katana-15-B13VGK:~/Documents/NCU/ReAct_Agent$ python main.py
============================================================
Task 2: Compare the main display specs of iPhone 15 and Samsung S24.
============================================================

[Step 1]
Thought: To compare the main display specs of iPhone 15 and Samsung S24, I first need to find the display specifications of the iPhone 15.
Action: Search
Action Input: iPhone 15 display specs
Observation: # Apple iPhone 15. | Dimensions | 147.6 x 71.6 x 7.8 mm |. ### Screen and keys. | Screen type | XDR OLED touch screen, HDR10, Dolby Vision, 16 million colours |. | Sc
reen size | 6.1 inches |. | Main camera | Dual 48 MP, f/1.6, 26mm (wide) + 12 MP, f/2.4, 13mm, 120˚ (ultra wide) |. | Front camera | Dual 12 MP, f/1.9, 23mm (wide) + SL 3D, (depth
/biometrics sensor) |. | Video recorder | 4K@24/25/30/60fps, 1080p@25/30/60/120/240fps, HDR, Dolby Vision HDR (up to 60fps), Cinematic mode (4K@30fps) |. | Chipset | Apple A16 Bio
nic (4 nm) |. | CPU | Hexa-core (2x3.46 GHz Everest + 4x2.02 GHz Sawtooth) |. | GPU | Apple GPU (5-core graphics) |. | 3.5 mm jack | No |. ### Connectivity. | Wi-Fi | 802.11 a/b/g
/n/ac/6, dual-band |. | Battery type | Non-removable Li-Ion 3349 mAh battery |. Excellent Very Good Good Fair Low No Connection. Excellent Very Good Good Fair Low No Connection.
---
*   [(Top)](https://en.wikipedia.org/wiki/IPhone_15#). *   [1 History](https://en.wikipedia.org/wiki/IPhone_15#History). *   [2 Design](https://en.wikipedia.org/wiki/IPhone_15#Des
ign). *   [3.1 Display](https://en.wikipedia.org/wiki/IPhone_15#Display). *   [3.3 Video output](https://en.wikipedia.org/wiki/IPhone_15#Video_output). *   [3.4 Battery](https://e
n.wikipedia.org/wiki/IPhone_15#Battery). *   [4 Software](https://en.wikipedia.org/wiki/IPhone_15#Software). *   [5 Specifications](https://en.wikipedia.org/wiki/IPhone_15#Specifi
cations). *   [6.1 Overheating](https://en.wikipedia.org/wiki/IPhone_15#Overheating). *   [7 See also](https://en.wikipedia.org/wiki/IPhone_15#See_also). *   [8 References](https:
//en.wikipedia.org/wiki/IPhone_15#References). *   [Read](https://en.wikipedia.org/wiki/IPhone_15). *   [Read](https://en.wikipedia.org/wiki/IPhone_15). D: 7.8 mm (0.31 in)[[1]](h
ttps://en.wikipedia.org/wiki/IPhone_15#cite_note-CNNInfo-1). *   **15 Plus**: 201 g (7.1 oz)[[1]](https://en.wikipedia.org/wiki/IPhone_15#cite_note-CNNInfo-1). Rear camera*   48 M
P, f/1.6, 26 mm equiv.[[A 1]](https://en.wikipedia.org/wiki/IPhone_15#cite_note-7) (wide). *   [4](https://en.wikipedia.org/wiki/IPhone_4 "IPhone 4"). *   [4s](https://en.wikipedi
a.org/wiki/IPhone_4s "IPhone 4s"). *   [5](https://en.wikipedia.org/wiki/IPhone_5 "IPhone 5"). *   [5s](https://en.wikipedia.org/wiki/IPhone_5s "IPhone 5s"). *   [6 / 6 Plus](http
s://en.wikipedia.org/wiki/IPhone_6 "IPhone 6"). *   [6s / 6s Plus](https://en.wikipedia.org/wiki/IPhone_6s "IPhone 6s"). *   [7 / 7 Plus](https://en.wikipedia.org/wiki/IPhone_7 "I
Phone 7"). *   [8 / 8 Plus](https://en.wikipedia.org/wiki/IPhone_8 "IPhone 8"). *   [X](https://en.wikipedia.org/wiki/IPhone_X "IPhone X"). *   [11](https://en.wikipedia.org/wiki/
IPhone_11 "IPhone 11"). *   [14 / 14 Plus](https://en.wikipedia.org/wiki/IPhone_14 "IPhone 14"). *   [15 / 15 Plus](https://en.wikipedia.org/wiki/IPhone_15). *   [16 / 16 Plus](ht
tps://en.wikipedia.org/wiki/IPhone_16 "IPhone 16"). *   [17](https://en.wikipedia.org/wiki/IPhone_17 "IPhone 17"). 6.   **[^](https://en.wikipedia.org/wiki/IPhone_15#cite_ref-6)**
Miller, Chance (September 13, 2023). **[^](https://en.wikipedia.org/wiki/IPhone_15#cite_ref-12)**Porter, Jon (September 23, 2021). **[^](https://en.wikipedia.org/wiki/IPhone_15#ci
te_ref-20)**Potuck, Michael (September 12, 2023). **[^](https://en.wikipedia.org/wiki/IPhone_15#cite_ref-27)**Apple Inc. **[^](https://en.wikipedia.org/wiki/IPhone_15#cite_ref-29)
**Phelan, David (September 28, 2023).
---
PhoneArena | Phone News, Reviews, Deals and Specs. # Apple iPhone 15. | Features | Super Retina XDR, HDR support, Oleophobic coating, Scratch-resistant glass, Ambient light sensor
, Proximity sensor |. | Learn more on our PhoneArena Display Test Results page. | Learn more on our PhoneArena Battery Test Results page. | Main camera | 48 MP (Sensor-shift OIS,
PDAF) Sensor name: Sony IMX904 Aperture size: F1.6 Focal length: 26 mm Pixel size: 2.0 µm |. | Second camera | 12 MP (Ultra-wide) Sensor name: Sony IMX772 Aperture size: F2.4 Foca
l Length: 13 mm |. | Learn more on our PhoneArena Camera Test Results page. | 3G Bands | 1, 2, 4, 5, 8 |. Discover the top rivals of Apple iPhone 15. Your iPhone 16 or older model
 is now worth less as Apple cuts trade-in values. Samsung's new return strategy is so unusual it could tempt even happy Galaxy S26 Ultra users to send their phones back. PhoneAren
a | Phone News, Reviews, Deals and Specs.
```

(a) Step 1 — Agent searches for iPhone 15 display specifications

```
(agentic) mtoan65@mtoan65-Katana-15-B13VGK:~/Documents/NCU/ReAct_Agent$ python main.py

[Step 2]
Thought: I have found the display specs of the iPhone 15, which include a 6.1-inch XDR OLED touch screen with HDR10, Dolby Vision, and 16 million colors. Now, I need to find the d
isplay specs of the Samsung S24 to compare them.
Action: Search
Action Input: Samsung S24 display specs
Observation: # Samsung S24 series tech specs (S24, S24+, S24 FE, S24 Ultra). We're here to give you more information on the Samsung S24 series release date, camera specs, and over
all features. Under the hood, the Galaxy S24+ is powered by a Qualcomm Snapdragon 8 Gen 3 processor, and other specs include a triple camera setup and a 4,900mAh battery. ## Samsu
ng Galaxy S24 FE specs. The Samsung Galaxy S24 FE introduces a 6.7-inch Dynamic AMOLED display with a smooth 120Hz refresh rate, with the Exynos 2400e processor under the hood. ##
 Samsung Galaxy S24 Ultra specs. The Samsung Galaxy S24 Ultra sits at the top of the pyramid with a 6.8-inch Dynamic AMOLED display with a 120Hz refresh rate. The Samsung S24 Ultr
a features a 5,000mAh battery, and its camera specs include a quad camera setup on the back. + **Video capture:** 3840x2160 (4K UHD) (60 fps). You can purchase a Samsung S24, S24+
, S24 FE, or S24 Ultra new or refurbished.
---
Size, 6.2-inch, 90.87% screen-to-body ; Type, Dynamic AMOLED, 120Hz, HDR, 2600 nits ; Resolution, 2340x1080px, 19.5:9 ratio, 416 PPI ; Protection
---
# Samsung Galaxy S24 (Exynos) Display test. We put the Samsung Galaxy S24 through our rigorous DXOMARK Display test suite to measure its performance across six criteria. The Samsu
ng Galaxy S24 nearly matched the display performance of the top-of-the-line S24 Ultra, thanks to achieving two top scores in readability and touch, but the smaller device was held
 back due to its management of artifacts. ***About DXOMARK Display tests**: For scoring and analysis in our smartphone and other display reviews, DXOMARK engineers perform a varie
ty of objective and perceptual tests under controlled lab and real-life conditions.* *Note that we evaluate display attributes using only the device's built-in display hardware an
d its still image (gallery) and video apps at their default settings.* *(For in-depth information about how we evaluate smartphone and other displays, check out our articles, "How
 DXOMARK tests display quality" and "A closer look at DXOMARK Display testing."*. Samsung Galaxy S24, color fidelity at 1000 lux in the Display-P3 color space.

[Step 3]
Thought: I have found the display specs of the Samsung S24, which include a 6.2-inch Dynamic AMOLED display with a 120Hz refresh rate, HDR, and 2600 nits. Now, I can compare the d
isplay specs of the iPhone 15 and the Samsung S24. The iPhone 15 has a 6.1-inch XDR OLED touch screen with HDR10, Dolby Vision, and 16 million colors, while the Samsung S24 has a
6.2-inch Dynamic AMOLED display with a 120Hz refresh rate, HDR, and 2600 nits.

Action: Search
Action Input: compare iPhone 15 and Samsung S24 display specs
Observation: # Apple iPhone 15 vs Samsung Galaxy S24 Specs Comparison. | 48 MP (Sensor-shift OIS, PDAF) Sensor name: Sony IMX904 Aperture size: F1.6 Focal length: 26 mm Pixel size
: 2.0 µm | 50 MP (OIS, PDAF) Sensor name: Samsung GN3 Aperture size: F1.8 Focal length: 24 mm Sensor size: 1/1.56" Pixel size: 1.0 µm |. | 12 MP (Ultra-wide) Sensor name: Sony IMX
772 Aperture size: F2.4 Focal Length: 13 mm 12 MP (Ultra-wide) Sensor name: Sony IMX564 Aperture size: F2.0 Focal Length: 13 mm Sensor size: 1/2.55" Pixel size: 1.4 µm | |. |  | 1
0 MP (Telephoto) Sensor name: Samsung S5K3K1 Aperture size: F2.4 Focal Length: 67 mm Sensor size: 1/3.94" Pixel size: 1.0 µm |. | | Full Specs | | --- | | Apple iPhone 15 Specs |
| PA Review | | Apple iPhone 15 Review | | | Full Specs | | --- | | Samsung Galaxy S24 Specs | | PA Review | | Samsung Galaxy S24 Review | |.
---
Samsung sells three Galaxy S24 devices, while Apple offers four iPhone 15 models. Similarly, Apple's top-of-the-line iPhone 15 Pro Max costs $100 less than the Galaxy S24 Ultra. S
amsung's Galaxy S24+ measures 6.2 by 3.0 by 0.30 inches and weighs 6.95 ounces, making it smaller and lighter than the iPhone 15 Plus (6.33 by 3.06 by 0.31 inches, 7.09 ounces) an
d the iPhone 15 Pro Max (6.29 by 3.02 by 0.32 inches, 7.81 ounces). Left to right: Galaxy S24 Ultra, iPhone 15 Pro Max. The S24 has a 6.2-inch display with FHD+ resolution (2,340
by 1,080 pixels), while the 6.7-inch S24+ and 6.8-inch S24 Ultra phones both have QHD+ (3,120 by 1,440 pixels) displays. Left to right: Galaxy S24 Ultra, iPhone 15 Pro Max. The Pr
o models get three rear cameras: a 48MP f/1.78 main lens with OIS, a 12MP f/2.2 ultra-wide camera with a 120-degree field of view, and a telephoto option. Moreover, the Apple Watc
h and AirTag work only with iPhones, while the Galaxy Watch and SmartTag 2 perform best with Samsung phones.
---
For video, the S24 offers 8K recording, while the iPhone 15 excels in 4K Cinematic Mode with smooth focus transitions. Choosing between them is
```

(b) Steps 2 & 3 — Agent searches for Samsung S24 specs, then issues a direct comparison search

```
(agentic) mtoan65@mtoan65-Katana-15-B13VGK:~/Documents/NCU/ReAct_Agent$ python main.py
[Step 4]
Thought: I have found the necessary information to compare the display specs of the iPhone 15 and the Samsung S24. The iPhone 15 has a 6.1-inch XDR OLED touch screen with HDR10, D
olby Vision, and 16 million colors, while the Samsung S24 has a 6.2-inch Dynamic AMOLED display with a 120Hz refresh rate, HDR, and 2600 nits.

Final Answer: The iPhone 15 features a 6.1-inch XDR OLED touch screen with HDR10, Dolby Vision, and 16 million colors, whereas the Samsung S24 boasts a 6.2-inch Dynamic AMOLED dis
play with a 120Hz refresh rate, HDR, and 2600 nits.

>>> Final Answer: The iPhone 15 features a 6.1-inch XDR OLED touch screen with HDR10, Dolby Vision, and 16 million colors, whereas the Samsung S24 boasts a 6.2-inch Dynamic AMOLED
 display with a 120Hz refresh rate, HDR, and 2600 nits.
```

(c) Step 4 — Agent synthesises both results and emits the Final Answer

Figure 3: Task 2 full console trace: Technical Specificity

2.3 Task 3 — Resilience & Reflection Test

Question: *"Who is the CEO of the startup 'Morphic' AI search?"*

Analysis: Resilience and Reflection

Resilience and Reflection Analysis. As seen in Figure **??**, the agent had self-correcting behaviour, which is the *Agentic Workflow* pattern described by Andrew Ng. Specifically, In step 1, it still returned the results but of `"Morphic Holding Inc"` rather than a useful result. Thus, in the next step 2, it adjusted the query to `"Morphic AI search startup CEO"` and then got the trully relevant information. This demonstrates that the agent directly satisfies the rubric's Reflection criterion.

```
(agentic) mtoan65@mtoan65-Katana-15-B13VGK:~/Documents/NCU/ReAct_Agent$ python main.py
============================================================
Task 3: Who is the CEO of the startup 'Morphic' AI search?
============================================================

[Step 1]
Thought: To find the CEO of the startup 'Morphic' AI search, I need to search for the latest information available on the company and its leadership.
Action: Search
Action Input: Morphic AI search CEO
Observation: The acquisition, anticipated to finalize in the third quarter of 2024, is seen as a strategic move for Eli Lilly, potentially positioning the company to enter the fir
st-line ulcerative colitis market. Morphic's lead drug, MORF-057, currently under multiple phase 2 studies for the treatment of ulcerative colitis and Crohn's disease, is at the h
eart of this deal. [...] In other recent news, Morphic Holding has been the subject of several significant developments. Following the announcement of Eli Lilly (NYSE:LLY)'s plans
 to acquire Morphic in a $3.2 billion deal, both TD Cowen and RBC Capital have downgraded Morphic's stock. TD Cowen shifted its rating from Buy to Hold, and RBC Capital moved from
 Outperform to Sector Perform, both aligning their price targets with the acquisition price of $57 per share. [...] # Morphic holding CEO sells shares worth over $1.2 million

By

Company News

Published 07/10/2024, 05:37 PM

Morphic holding CEO sells shares worth over $1.2 million

View all comments (0)0

Morphic holding CEO sells shares worth over $1.2 million

Morphic holding CEO sells shares worth over $1.2 million

Morphic Holding, Inc. (NASDAQ:MORF) President and CEO, Praveen P. Tipirneni, has made significant transactions in the company's stock, according to a recent SEC filing. The execut
ive sold 21,582 shares of common stock on July 8, 2024, at prices ranging from $55.79 to $55.80, resulting in a total value of over $1.2 million.
---
Over the past year, the insider has sold a total of 54,330 shares and has not made any purchases of the company's stock.

The insider transaction history at Morphic Holding Inc reveals a pattern of insider activity. In the past year, there have been 3 insider buys and 23 insider sells.

Morphic Holding Inc CEO Praveen Tipirneni Sells 13,923 Shares

Morphic Holding Inc CEO Praveen Tipirneni Sells 13,923 Shares



On the date of the insider's recent transaction, shares of Morphic Holding Inc were trading at $27.49, resulting in a market cap of $1.351 billion.

The stock's price-to-GF-Value ratio stands at 59.76, with a GF Value of $0.46, indicating that Morphic Holding Inc is significantly overvalued according to GuruFocus' valuation me
trics. [...] Morphic Holding Inc CEO Praveen Tipirneni Sells 13,923 Shares

Morphic Holding Inc CEO Praveen Tipirneni Sells 13,923 Shares
```

(a) Step 1 (part 1/4) — Agent queries `Morphic AI search CEO`; observation returns irrelevant results about `Morphic Holding Inc.`

```
(agentic) mtoan65@mtoan65-Katana-15-B13VGK:~/Documents/NCU/ReAct_Agent$ python main.py
# Morphic Holding Inc CEO Praveen Tipirneni Sells 13,923 Shares

GuruFocus Research

1 min read

MORF

On January 16, 2024, CEO of Morphic Holding Inc (NASDAQ:MORF), Praveen Tipirneni, sold 13,923 shares of the company, according to a recent SEC Filing. Morphic Holding Inc is a bio
pharmaceutical company focused on the discovery and development of oral small-molecule integrin therapeutics for serious chronic diseases, including fibrosis, cancer, and autoimmu
ne conditions.

 Warning! GuruFocus has detected 2 Warning Signs with MORF.
---
Jaynti Kanani is the current CEO of Morphic. They are also angel investors in 19 startups, including Polygon.

Morphicis headquartered in San Francisco, United States.

Morphicis a funded company,having raised a total of  $\\\\\ in its 1st and only funding round.

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# Morphic - Company Profile

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```

(b) Step 1 (part 2/4) — Observation continues; `Jaynti Kanani` is mentioned as a possible CEO of a different `Morphic` entity

Figure 4: Task 3 console trace — Step 1 observation (parts 1–2)

```
(agentic) mtoan65@mtoan65-Katana-15-B13VGK:~/Documents/NCU/ReAct_Agent$ python main.py

## Morphic - About the company

Morphic is an unfunded company based in San Francisco (United States), foundedbyJaynti Kanani. It operates as a Morphic — A new era of storytelling. Morphic has not raised any fun
ding yet.

### Company Details

Morphic is transforming the future of storytelling using breakthrough technologies.

Website
:   morphic.com

Social [...] Website
:   morphic.com

Social

Key Metrics

Location

San Francisco, United States

Stage

Unfunded

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## Morphic's funding and investors

Morphic has not raised any funding rounds yet.

## Morphic's founders and board of directors

Founder? Claim Profile

The founders of MorphicisJaynti Kanani.

Here are the details of Morphic'skey team members:
```

(a) Step 1 (part 3/4) — Observation continues; company profile data for a storytelling startup named `Morphic`, founded by `Jaynti Kanani`

```
(agentic) mtoan65@mtoan65-Katana-15-B13VGK:~/Documents/NCU/ReAct_Agent$ python main.py
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## Morphic's funding and investors

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## Morphic's founders and board of directors

Founder? Claim Profile

The founders of MorphicisJaynti Kanani.

Here are the details of Morphic'skey team members:

 Jaynti Kanani: Founder & CEO of Morphic and founder of 1 other company. They are also an angel investor in 19 startups.

  Contact Info:1 email address

View details ofMorphic's Founder profiles and Board Members

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```

(b) Step 1 (part 4/4) — Observation concludes; `Jaynti Kanani` confirmed as Founder & CEO of this unrelated `Morphic` entity

Figure 5: Task 3 console trace — Step 1 observation (parts 3–4)

```
(agentic) mtoan65@mtoan65-Katana-15-B13VGK:~/Documents/NCU/ReAct_Agent$ python main.py

[Step 2]
Thought: The search results provided information about Morphic Holding Inc., which seems to be a different company from the 'Morphic' AI search startup I'm looking for. The CEO of
 Morphic Holding Inc. is Praveen P. Tipirneni, but this is not the company I need information on. I will try searching again with a different query to find the correct information
.

Action: Search
Action Input: Morphic AI search startup CEO
Observation: Today I am thrilled to welcome to the show Weston Baker, Founder & CEO of Morphic and an overall force in the startup ecosystem.
---
Weston Baker, Founder-CEO, Morphic https://www.linkedin ... startup studio that has worked as partners (both cash and equity)
---
Weston Baker, Founder & CEO of Morphic and an ... Think Bigger: How AI is Reshaping Staffing, IP Strategy, and R&D Funding for Startups.

[Step 3]
Thought: The search results provided information about the CEO of the Morphic AI search startup, which is Weston Baker. I have found the answer I need.

Final Answer: The CEO of the startup 'Morphic' AI search is Weston Baker.

>>> Final Answer: The CEO of the startup 'Morphic' AI search is Weston Baker.
```

Figure 6: Steps 2 & 3 — Agent reflects, reformulates the query to `Morphic AI search startup CEO`, retrieves **Weston Baker** as the correct answer, and emits the Final Answer

References

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